FLOODGAUGE · PRODUCT REQUIREMENTS AND SPECIFICATIONS

FloodGauge product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-31	Amish Chadha	Kit 1.9.1 upgrade: renamed from FLG-REQ-001 to the product requirements and specifications (PRS); every requirement renumbered PR-001 to PR-017 (old R numbers kept in the Was column), traced to the user requirements in FLG-URS-001, with tolerance and units; PR-018 and PR-019 added for UR-017 and UR-010; status from FLG-CAL-001 v0.7; PR-015 now at risk and PR-006 limited to SF9 and SF10 after the independent review
0.8	2026-10-02	Amish Chadha	Status from FLG-CAL-001 v0.6 (sizing.py rerun); R16 cost USD 205.50 with the tamper-resistant fixings priced
0.7	2026-10-02	Amish Chadha	R12 relaxed to 120 min for the pilot surface route (90 min at permanent sites), now met on paper; R15 fix named; first alert band 100 mm (decisions of 2026-10-02)

0.6	2026-10-01	Amish Chadha	Status from FLG-CAL-001 v0.4 (constructable design, FLG-DDR-003); R16 reported against the value-engineering target
0.5	2026-09-26	Amish Chadha	Budget approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from FLG-CAL-001; R16 restated to FloodGauge-specific parts under FLG-DDR-001 D1
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

FloodGauge product requirements and specifications

What the design must do, each requirement traced to a user need in FLG-URS-001. The targets are proposals for review with a co-design partner. Method: STANDARDS section 22. The product requirements were written as R1 to R17 in versions 0.1 to 0.8 of FLG-REQ-001 and are renumbered here as PR-001 to PR-017; the Was column keeps the old number so earlier documents and the calculation note still match.

These requirements are checked by calculation in FLG-CAL-001 v0.7, on the constructable design of FLG-DDR-003, which is concept option A in FLG-OPT-001 (the two other options, B and C, are scored there against the same user requirements). The design choices behind them were decided by Amish on 2026-09-25 (FLG-DDR-001 and FLG-DDR-002). Three targets changed at TRL 3: PR-016 covers FloodGauge-specific parts only, with the FieldNode core costed in its own repo (DDR-001 D1); PR-001 allows a mounting height of 2.5 to 5.0 m, set per site to the road authority's clearance rule; and PR-003 is restated to the reach of the drain head plus a drain-full signal (both DDR-002). On 2026-10-02 Amish relaxed PR-012 to 120 min for the pilot surface route, keeping 90 min for permanent sites (FLG-DEC-001, item 5). On 2026-09-26 Amish set `budget_usd` to $160 (DDR-002); on 2026-10-01 he described budgets as hypothetical value-engineering targets, so PR-016 is reported over or under the $160 target rather than as met or not met. The status column states "not met" plainly.

Table 1. Product requirements and specifications at TRL 3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001, UR-009	Performance	Street depth range and mounting height	Measure water depth on the street at the gauge point, with the head mounted at the height the road authority's clearance rule sets for the site	0 to 600 depth; head 2,500 to 5,000 above the road	range	mm	Range calculation against sensor data (CAL), TRL 3	Met on paper: reference lens at 4.6 m clears a 4.3 m vehicle; range 1.9 to 5.0 m over the band, radar to 20 m	R1
PR-002	UR-001, UR-003	Performance	Street depth accuracy	Street depth error over -20 to 50 °C air, still water	10	max, plus or minus	mm	Error budget (CAL), TRL 3; bench test over a tank at TRL 4	Met on paper with radar: 6.5 mm root sum square (12.0 mm worst-case sum), module accuracy assumed. **Not met** by the ultrasonic variant (27.0 mm compensated at 4.6 m)	R2
PR-003	UR-002	Performance	Catch basin level range	Continuous reading from 50 mm above the basin floor to 330 mm below the road, plus a drain-full signal above that	1,250 to 330 below road; drain-full trip at 310	range	mm	Geometry of stilling tube and head (CAL), TRL 3	Met on paper: continuous from 1,250 to 330 mm below the road; wet probe trips at 310 mm	R3
PR-004	UR-002, UR-003	Performance	Drain level accuracy	Catch basin level error	20	max, plus or minus	mm	Error budget (CAL), TRL 3	Met on paper: 11.0 mm with the NTC in the head (19.7 mm without)	R4
PR-005	UR-005	Performance	Sampling rate	Sample every 60 s		max interval	s	Sampling logic review	Met by design	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				normally; every 10 s when level is above the first alert band or rising faster than 20 mm/min	60 normal; 10 in an event			(CAL), TRL 3		
PR-006	UR-004	Performance	Alert latency	Time from threshold crossing to a message at the alert service, 95 % of events	120	max	s	Latency simulation (CAL), TRL 3	Met on paper at SF9 and SF10 on a private gateway: 82 s at the 95th percentile (85 s on a city server); 110 s worst case without a lost packet. Not met at SF11 (123 s) and SF12 (201 s) in event mode (independent review)	R6
PR-007	UR-013	Electrical	Radio airtime within network rules	Within the EU868 1 % duty cycle always; within The Things Network fair use (30 s uplink airtime per day) in normal mode	1 % duty; 30 s/day	max	% and s/day	Airtime calculation (CAL), TRL 3	**At risk:** normal mode 23.7 s/day at SF9; event mode within 1 % at SF7 to SF10 but not at SF11 or SF12 unless reporting stretches to 99 s or 181 s; normal mode exceeds fair use at SF10 and slower	R7
PR-008	UR-006	Electrical	Energy autonomy	Run on stored energy with no sun at	5	min	days	Power budget against FieldNode	Met on paper: 68 days nominal, 47	R8

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				event-mode sampling				(CAL), TRL 3	days at -20 °C; 9.3 % of the FieldNode 100 mW allowance	
PR-009	UR-007	Materials	Survive submersion of the drain head	IP68 to 2 m for 72 h in stormwater with silt	IP68, 2 m, 72 h	min	rating, m, h	Enclosure design review (CAL), TRL 3; immersion test at TRL 4	**At risk:** electronics potted, but the transducer face keeps the module's IP67 seal	R9
PR-010	UR-011	Software	Reject false readings	No alert caused by a vehicle, person or debris under the street head	1 false alert per gauge per year	max	alerts/ year	Plausibility logic review (CAL), TRL 3; field log at TRL 4	To verify at TRL 4: rules defined, including a recorded dry-road background against the curb echo	R10
PR-011	UR-012	Software	Privacy	The device records levels, status and battery only; no camera or microphone; no image or audio leaves the device	0 cameras and microphones	exact	count	Design review (BOM, PRC), TRL 3	Met by design	R11
PR-012	UR-008	Usability	Installation	Two-person crew, from the surface; no road drilling; no entry into the catch basin. 120 min on the pilot surface cable route (relaxed on 2026-10-02, FLG-	120 pilot; 90 permanent	max	min	Installation sequence review (CAL, BLD), TRL 3	Met on paper for the pilot route: 120 min, with no civil work (DDR-002)	R12

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				DEC-001 item 5); 90 min at permanent sites built on the conduit route with road works						
PR-013	UR-014	Performance	Level datum	Head height surveyed to the road surface at the gauge point; checked yearly against the depth marker	5	max, plus or minus	mm	Survey method (CAL), TRL 3	Met on paper: tape survey plus a recorded dry-road radar reading	R13
PR-014	UR-006, UR-007	Materials	Operating environment	Work in air from -20 to 50 °C with UV, rain and road spray; FieldNode charges only from 0 to 45 °C	-20 to 50	range	°C	Material and cell review (CAL), TRL 3	**At risk:** inherits FieldNode's interior heat finding (FND R2, R3); the drain head module is rated -15 to 60 °C	R14
PR-015	UR-009, UR-010	Physical architecture	Mounting and mass on the pole	Fit 40 to 60 mm poles with band clamps; total added mass on the pole	5.0	max	kg	Mass from the CAD model (CAL), TRL 3; weigh at TRL 4	**At risk:** 4.994 kg (margin 0.006 kg, inside the accuracy of an estimate from model volumes); twist factor 1.8 with the M8 A4-80 anti-rotation through-bolt, 1.45 on friction alone on a 60 mm pole and	R15

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
									0.97 on a 40 mm pole	
PR-016	UR-016	Cost	Cost	FloodGauge-specific parts at quantity 1; FieldNode costed separately (FND-CAL-001)	160	max	USD	Priced BOM (bom/bom.csv), TRL 3	Over target by $45.50: $205.50 for the constructable design ($344.50 with the $139.00 FieldNode core)	R16
PR-017	UR-015	Software	Open data	Levels published in an open, documented format through the gateway	JSON or CSV	exact	format	Design review (PRC), TRL 3	Met by design	R17
PR-018	UR-017	Safety	Tamper resistance	Every fixing within reach from the sidewalk needs a security tool: security (snake-eye) nuts on the eight masonry anchors and pin-Torx security screws on the marker band clamps	10 of 10 fixings within reach	min	count	CAD model and BOM review (CAL, BLD), TRL 3	Met by design (new in this version; the fixings were added in FLG-DDR-003)	
PR-019	UR-010	Safety	Cover profile and edges	Surface cable covers project no more than the hat-section height, and every exposed edge of a	16	max	mm	CAD model (cad_check edge breaks) and drawing FLG-DWG-001, TRL 3	Met by design: hat section 16 mm high with 2 mm sheet; edges broken in the model	

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				made part is broken						

Status values: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The road surface at the gauge point is the datum for street depth. Depth equals the surveyed head height minus the measured range.
- "Event" means water above the first alert band or rising faster than 20 mm/min.
- FieldNode figures (usable energy, allowance, airtime per uplink, mass, cost, charge temperature window) come from FND-CAL-001 in the FieldNode repo.
- Alert depths of 150 mm and 300 mm follow NWS guidance on water that can knock over an adult and carry away a car (NWS). They are defaults under FLG-DDR-001 D6, to be set locally with the partner city. Under FLG-DDR-002 the alert service also supports a first band below 150 mm; Amish set it at 100 mm on 2026-10-02 unless the partner's own practice says otherwise (FLG-DEC-001, item 6).
- The remaining assumptions for every status above are in FLG-CAL-001, Table 1.

Requirements not met or at risk

- **PR-012 (was R12):** the surface cable route (DDR-002) removes the coring and trenching, but fitting its cover takes the installation to 120 min. Amish relaxed it to 120 min for the pilot route on 2026-10-02 (FLG-DEC-001, item 5), so it is met on paper; 90 min stays the target at permanent sites.
- **PR-007 at risk:** slow spreading factors need a longer event interval; event reporting belongs on a private or city gateway (D7).
- **PR-009 at risk:** the transducer face seal decides immersion survival.
- **PR-014 at risk:** interior heat in FieldNode, and the drain module's -15 °C lower rating.
- **PR-002** would not be met if the ultrasonic street variant were used.
- **PR-015 at risk:** 4.994 kg against 5.0 kg on masses taken from the model volumes after the parts added for construction (FLG-DDR-003); the margin of 6 g is inside the accuracy of the estimate (independent review), so it is no longer called met on paper; weigh at TRL 4. The twist factor of 37 in earlier versions was wrong (FLG-DDR-004 F2): it is 1.8 on the bolt, and friction alone gives only 0.97 on a 40 mm pole.
- **PR-006 limited to SF9 and SF10:** event-mode alerts take 123 s at SF11 and 201 s at SF12 (FLG-CAL-001 F4). Decided 2026-10-02 (FLG-DEC-001, item 3): if the weighed mass is over 5.0 kg, the bracket plate goes to 2.5 mm.
- **PR-016:** the $160 is a value-engineering target (Amish, 2026-10-01), not a limit; the constructable design is estimated at $205.50, $45.50 over it.
- **PR-010:** cannot be verified before a field log exists (TRL 4).